SUPPLEMENTARY MATERIAL OF

LOAD: Load-Aware Dual-Price Routing for Cross-Layer Secure Relaying in Multi-Layer NTNs

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APPENDIX A
NOTATIONS AND VARIABLES

All variables in the paper can be summarized in Table III.

TABLE III
SUMMARY OF NOTATIONS

Symbol	Description	Symbol	Description
Sets and Indices		**Performance Metrics**	
$\mathcal{M}$	Set of indices for Eves.	$\mathrm{SNR}^{\mathsf{s}}_{(i,j)}$	SNR of the legitimate link (i,j).
$\mathcal{I}$	Set of indices for network nodes, $\{0,1,...,I\}$.	$\mathrm{SNR}^{\mathsf{e}}_{(i,e)}$	SNR of the wiretap link (i,e).
$\mathcal{U}$	Set of indices for users, $\{1,...,U\}$.	$C_{(i,j)}$	Secrecy capacity of the link (i,j).
$\mathcal{N}$	Set of nodes and users in the graph $\mathcal{G}$.	$\mathbb{P}_{(i,j)}$	SPSC Probability for the link (i,j).
$\mathcal{E}$	Set of edges in the graph $\mathcal{G}$.	$\gamma_{(i,j)}$	Spectral efficiency of the link (i,j).
$\mathcal{E}_u$	Set of edges that constitute the relay path for user u.	η_u	Relay throughput for user u.
S	A subset of nodes for the spanning tree constraint.	h_u	Number of hops in the relay path for user u.
Physical Layer Parameters		**Graph and Routing Variables**	
λ_i	Density of Eves in the network layer i.	$\mathcal{G}_{\text{all}}$	Directed network routing graph.
$\boldsymbol{p}_i, \boldsymbol{p}_e$	Position vector of node i and Eve e.	$x_{(i,j)}$	Binary variable indicating if edge (i,j) is in the graph $\mathcal{G}$.
$d^{\mathsf{s}}_{(i,j)}$	Distance between node i and node j.	$x_{(i,j),u}$	Binary variable indicating if edge (i,j) is in the relay path $\mathcal{E}_u$ for user u.
$d^{\mathsf{e}}_{(i,e)}$	Distance between node i and Eve e.		
α_i	Path loss exponent for the link from node i.		
$h^{\mathsf{s}}_{(i,j)}, h^{\mathsf{e}}_{(i,e)}$	Small-scale fading channel coefficient for the legitimate link (i,j) and the wiretap link (i,e).		
ρ_i	Transmit power density of node i.		
σ_i	Jamming power density of node i.		
n_0	Noise power spectral density.		
$P_i^{\max}, P_i^{\min}$	Max/min transmission power of a node i.		
$\beta_{(i,j),u}$	Bandwidth allocated to user u on the link (i,j).		

APPENDIX B
APPROXIMATION OF ERGODIC SPECTRAL EFFICIENCY

We approximate the ergodic spectral efficiency to capture long-term network throughput. By Jensen's inequality, this approximation provides an upper bound on the ergodic spectral efficiency. Figure 12 validates the approximation by comparing the ergodic spectral efficiency with its no-fading counterpart. Over SNRs in the range [-40, 30] dB, the approximation closely matches the true curve in the low-SNR regime, and the gap widens as SNR increases. This gap can be reduced by introducing a scaling factor α that multiplies (9). A least-squares fit yields $\alpha = 0.8908$, achieving a mean squared error (MSE) of 0.008 over the specified SNR range. Hence, the approximation in (9) does not materially affect the decision variables, and any residual bias can be compensated by this simple linear rescaling.

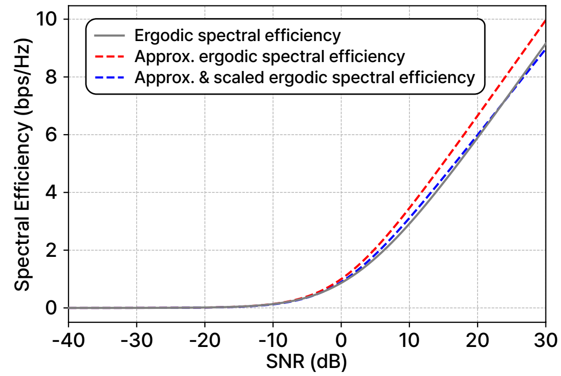

Fig. 12. Comparison between ergodic spectral efficiency and its approximation

Appendix C
Derivation of SPSC Approximation

Using $|h^{\mathsf{e}}_{(i,e)}|^2 \sim \mathsf{Exp}(1)$, we condition on $z = |h^{\mathsf{s}}_{(i,j)}|^2$. For $\sigma_i > 0$, define

$$z_\tau = \frac{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}{\sigma_i G_{(i,j)}}.$$

The SPSC condition is satisfied for every eavesdropper when $z \geq z_\tau$. For $0 < z < z_\tau$, the probability that an eavesdropper at $r = d^{\mathsf{e}}_{(i,e)}$ violates the SPSC condition is

$$\mathbb{P}\Big(\mathrm{SNR}^{\mathsf{e}}_{(i,e)} > \mathrm{SNR}^{\mathsf{s}}_{(i,j)} \mid z, r\Big) = \exp\left[-\frac{z n_0 r^{\alpha_i}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0 - \sigma_i G_{(i,j)} z}\right],$$

With

$$\kappa_i = \lambda_i \frac{2\pi}{\alpha_i} \Gamma\Big(\frac{2}{\alpha_i}\Big),$$

the HPPP probability generating functional and polar integration yield

$$\lambda_i \int_{\mathbb{R}^2} \exp\left[-\frac{z n_0 r^{\alpha_i}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0 - \sigma_i G_{(i,j)} z}\right] d\boldsymbol{p}_e = \kappa_i (d^{\mathsf{s}}_{(i,j)})^2 \left[\frac{1}{z} - \frac{\sigma_i G_{(i,j)}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}\right]^{\frac{2}{\alpha_i}}. \tag{48}$$

Accordingly, the exact SPSC probability is

$$\mathbb{P}_{(i,j)} = e^{-z_\tau} + \int_0^{z_\tau} \exp\left[-z - \kappa_i (d^{\mathsf{s}}_{(i,j)})^2 \left(\frac{1}{z} - \frac{\sigma_i G_{(i,j)}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}\right)^{\frac{2}{\alpha_i}}\right] dz. \tag{49}$$

By taking

$$x = \kappa_i (d^{\mathsf{s}}_{(i,j)})^2 \left(\frac{1}{z} - \frac{\sigma_i G_{(i,j)}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}\right)^{\frac{2}{\alpha_i}}. \tag{50}$$

and applying integration by parts, the automatic-success term e^{-z_τ} is canceled, and the SPSC probability is written as

$$\mathbb{P}_{(i,j)}(\sigma_i) = \int_0^\infty \exp\left[-x - \left(\frac{\sigma_i G_{(i,j)}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0} + \left(\frac{x}{\kappa_i (d^{\mathsf{s}}_{(i,j)})^2}\right)^{\frac{\alpha_i}{2}}\right)^{-1}\right] dx. \tag{51}$$

Because e^{-x} makes $x = O(1)$ the dominant region of the integral, we consider the operating condition

$$\big[\kappa_i (d^{\mathsf{s}}_{(i,j)})^2\big]^{-\alpha_i/2} \ll \frac{\sigma_i G_{(i,j)}}{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}, \tag{52}$$

Under this condition, the first-order expansion gives

$$\mathbb{P}_{(i,j)} = \exp\left[-\frac{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}{\sigma_i G_{(i,j)}}\right]\Bigg[1+\Gamma\left(1+\frac{\alpha_i}{2}\right)\left[\kappa_i (d^{\mathsf{s}}_{(i,j)})^2\right]^{-\frac{\alpha_i}{2}}\left(\frac{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i} n_0}{\sigma_i G_{(i,j)}}\right)^2 + O\Big(\left[\kappa_i (d^{\mathsf{s}}_{(i,j)})^2\right]^{-\alpha_i}\Big)\Bigg]. \tag{53}$$

Finally, re-exponentiating the first-order correction gives the closed-form approximation in (18),

$$\widehat{\mathbb{P}}_{(i,j)}(\sigma_i) = \exp\left[-\frac{(d^{\mathsf{s}}_{(i,j)})^{\alpha_i}}{\frac{\sigma_i G_{(i,j)}}{n_0} + \Gamma\left(1+\frac{\alpha_i}{2}\right)\kappa_i^{-\alpha_i/2}}\right]. \tag{54}$$

Appendix D
Proof of Optimality of the Fixed-Topology RRM

Theorem 1 (Optimality of RRM). *Let $\mathcal{G}$ be a fixed feasible network topology. Then, the solution $(\beta^*_{(i,j),u}, \rho^*_i, \sigma^*_i)$ obtained by solving $\mathcal{P}_1$ under the given $\mathcal{G}$ is globally optimal for the fixed-topology RRM subproblem, provided that $P_i^{\max} - \tau_i \geq P_i^{\min}$ for transmitters with outgoing routed links.*

Proof. Fix a feasible routing graph $\mathcal{G}$. Then the routing indicators $x_{(i,j),u}$, hop counts h_u, and link distances $d^{\mathsf{s}}_{(i,j)}$ are constants. The remaining decision variables are the bandwidth allocation $\beta_{(i,j),u}$ and the power split (ρ_i, σ_i).

We first fix (ρ_i, σ_i). Then each $\gamma_{(i,j)}$ is a positive constant, and the throughput constraint in $\mathcal{P}_2$ is equivalent to

$$\beta_{(i,j),u} \geq \frac{x_{(i,j),u} h_u}{\gamma_{(i,j)}}\eta. \tag{55}$$

Summing (55) over the outgoing links and users of any transmitter i that carries routed flow gives

$$\eta \sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}} \frac{x_{(i,j),u} h_u}{\gamma_{(i,j)}} \leq \sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}} \beta_{(i,j),u} \leq B. \tag{56}$$

Hence any feasible allocation must satisfy

$$\eta \leq \min_{\substack{i\in\mathcal{I}:\\ \sum_{j,u} x_{(i,j),u}>0}} B\left(\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}} \frac{x_{(i,j),u} h_u}{\gamma_{(i,j)}}\right)^{-1}. \tag{57}$$

This is an upper bound on the common throughput for the fixed powers. The allocation in (22) attains this bound: for every used outgoing link of transmitter i,

$$\frac{\beta^*_{(i,j),u}\gamma_{(i,j)}}{h_u} = B\left(\sum_{j\in\mathcal{N}}\sum_{u\in\mathcal{U}} \frac{x_{(i,j),u} h_u}{\gamma_{(i,j)}}\right)^{-1}. \tag{58}$$

Therefore, (22) is a globally optimal bandwidth allocation for any fixed feasible power split.

It remains to optimize the power split. For a fixed graph, the SPSC approximation in (18) is strictly increasing in σ_i. Thus, the SPSC constraints reduce to $\sigma_i \geq \tau_i$, where τ_i is defined in (23). The theorem assumes feasibility, i.e., $P_i^{\max} - \tau_i \geq P_i^{\min}$ for every transmitter that carries routed flow.

The remaining bandwidth upper bound in (57) depends on the power split through the spectral efficiencies. Since

$$\gamma_{(i,j)} = \log_2\left(1 + \mathrm{SNR}^{\mathsf{s}}_{(i,j)}\right) \tag{59}$$

is monotone increasing in the data-transmission power density ρ_i, increasing ρ_i cannot decrease the best fixed-graph throughput. For any feasible point with $\sigma_i > \tau_i$, the point (ρ_i, σ_i) is dominated by $(P_i^{\max} - \tau_i, \tau_i)$ because the latter preserves the SPSC constraint and provides weakly larger data-transmission power. If $\sigma_i = \tau_i$ but $\rho_i < P_i^{\max} - \tau_i$, increasing ρ_i also preserves feasibility and cannot reduce throughput. Hence an optimal power split is

$$\sigma_i^* = \tau_i, \qquad \rho_i^* = P_i^{\max} - \tau_i, \tag{60}$$

which is (26).

Combining the two parts, (26) maximizes the fixed-graph throughput bound induced by the bandwidth allocation subproblem, and (22) attains that bound for the resulting spectral efficiencies. Thus, $(\beta^*_{(i,j),u}, \rho^*_i, \sigma^*_i)$ globally solves this RRM subproblem. $\square$

APPENDIX E
TRAINING DETAILS OF LEARNED ROUTING POLICIES

The learned routing schemes are trained offline from candidate paths generated on the SPSC-feasible graph. The supervision is produced by the analytical fixed-topology RRM evaluation used throughout the paper; hence the models are optimization-informed learning-to-rank policies, not reinforcement-learning controllers and not direct throughput regressors.

A. Training Protocol

The reported checkpoints use 128 training topology–condition instances spanning the SPSC-threshold and Eve-density regimes evaluated in the paper. The condition generator uses seed 20260622 for LOAD and seed 0 for the edge-cost models, while parameter initialization and sample shuffling use seed 0 for all three models. The sequential candidate procedure produces fixed training sets of 5,686 routing states for LOAD, 6,087 for the edge-cost MLP, and 6,088 for the edge-cost GNN. Each state forms one Adam update, and the final checkpoint after 30 epochs is used. An additional 100 topology–condition instances with disjoint topology indices are reserved for post-training evaluation and are not used for gradient updates or checkpoint selection. Accordingly, Fig. 5 reports per-epoch averages over the training samples.

B. Candidate-Path Supervision

Each training sample corresponds to a partial routing tree $\mathcal{G}_s$ and one user whose route is to be selected next. A finite candidate set is generated under the SPSC and hop constraints, and every candidate is evaluated by temporarily inserting the path into $\mathcal{G}_s$ and recomputing the corresponding RRM-induced quality. The selected teacher candidate is then committed before the next user is processed, so the training samples follow the same sequential routing state distribution used at inference.

For LOAD, candidate quality is defined by the post-decision bottleneck relay load in (46). For the edge-cost MLP/GNN baselines, candidate quality is defined by the post-decision max-min throughput obtained after inserting each candidate path. Thus, all learned schemes are trained from optimization-generated labels over feasible candidate paths rather than from externally annotated routes. The edge-cost candidate pools combine the minimum-hop and minimum-distance paths with paths from 100 randomized shortest-path predecessor maps.

C. DP-GNN Architecture and Training Objective

LOAD is a relay-level model. Its 14-dimensional relay-state vector contains a four-dimensional relay-type encoding, log-scaled load and RRM denominator, normalized inverse throughput, a baseline dual prior, log-scaled child count, farthest outgoing child distance, downstream hop mass, bandwidth, power budget, and path-loss exponent. The relay graph used for message passing contains SPSC-feasible relay-to-relay edges with edge features given by log-distance, SPSC slack, and a binary indicator of whether the edge is already present in the partial tree. In the reported model, five future-demand features are pooled from approximate paths of currently unconnected users and injected before message passing.

The architecture first embeds the future-demand features, performs graph message passing over the feasible relay graph, and concatenates the resulting relay embedding with the relay-state feature vector. A dual head then outputs one scalar logit s_i^θ per relay. The resulting softmax distribution is the prospective relay price $\boldsymbol{\pi}^\theta$ in (38), which is the only learned quantity used by the decoder. Candidate reduced edge costs are computed analytically from (39), and path costs are obtained by summing these costs as in (40).

Before the softmax operations, the predicted path costs are standardized by their candidate-set mean and standard deviation, while the bottleneck targets are normalized by the best-to-worst range. The implemented LOAD loss follows the quality-KL objective in (47), with the entropy term normalized by $\log K$ for a candidate set of size K:

$$\mathcal{J}_{\mathrm{DP}}(\theta) = D_{\mathrm{KL}}\big(\mathbf{q} \,\|\, \mathbf{p}^\theta\big) - \zeta \frac{H(\mathbf{p}^\theta)}{\log K} + \lambda_{\mathrm{dual}} D_{\mathrm{KL}}\big(\boldsymbol{\pi}^{\mathrm{base}} \,\|\, \boldsymbol{\pi}^\theta\big)\,. \tag{61}$$

Here, $\boldsymbol{\pi}^{\mathrm{base}} = \mathrm{softmax}(\alpha_{\mathrm{base}}\widetilde{\boldsymbol{L}})$, where $\widetilde{\boldsymbol{L}}$ is the standardized current relay load. The first term aligns reduced-cost ranking with the target distribution induced by post-decision bottleneck loads, while the dual KL term anchors the learned prices to this reference distribution. The optional entropy term can discourage early collapse to a single candidate path. The reported LOAD checkpoint uses $\alpha_{\mathrm{base}} = 0$ and $\zeta = 0$, so $\boldsymbol{\pi}^{\mathrm{base}}$ is uniform and the entropy term is inactive. The future-demand GNN has hidden dimension 128, five message-passing layers, and a dual head with two 128-dimensional hidden layers. Adam uses learning rate $3{\cdot}10^{-4}$, $T = 1$, $T_q = 1$, $\lambda_{\mathrm{dual}} = 0.02$, maximum hop 24, hop slack 12, and gradient clipping at norm 5. The model contains 449,793 trainable parameters.

D. *Edge-Cost MLP/GNN*

The edge-cost baselines learn a positive edge metric rather than relay dual prices. For each feasible directed edge, the input edge features include endpoint types, whether the child is the current target user, source/target distances, current load and pressure at both endpoints, child counts, current-tree indicators, path-loss exponent, bandwidth, and SPSC slack. The MLP maps this 27-dimensional edge feature vector independently to a positive scalar edge cost using a softplus output. The edge-cost GNN uses the same edge features together with 13-dimensional node features, performs message passing over the feasible graph, and predicts a positive edge cost from the source embedding, destination embedding, and edge embedding.

For a candidate path P_k, the learned path cost is the sum of its edge costs, denoted by C_k^{ϕ}. The path preference is

$$p_k^{\phi} = \frac{\exp(-\widetilde{C}_k^{\phi}/T)}{\sum_{\ell=1}^{K} \exp(-\widetilde{C}_\ell^{\phi}/T)}, \tag{62}$$

where $\widetilde{C}_k^{\phi} = (C_k^{\phi} - \bar{C}^{\phi})/\max(s_C, 10^{-3})$, and $\bar{C}^{\phi}$ and s_C are the mean and standard deviation over the candidate set. Let S_k be the post-decision throughput score of candidate P_k, and let $k^{\star} = \arg\max_k S_k$. The throughput-regret target is formed as

$$r_k^{\text{edge}} = \frac{S_{k^\star} - S_k}{S_{k^\star} - \min_\ell S_\ell + \epsilon}, \qquad q_k^{\text{edge}} = \frac{\exp(-r_k^{\text{edge}}/T_q)}{\sum_{\ell=1}^{K} \exp(-r_\ell^{\text{edge}}/T_q)}. \tag{63}$$

The edge-cost training objective is

$$\mathcal{J}_{\text{edge}}(\phi) = D_{\text{KL}}\big(\mathbf{q}^{\text{edge}} \,\|\, \mathbf{p}^{\phi}\big) - w_{\text{ce}} \log p_{k^\star}^{\phi} + w_{\text{m}} \frac{1}{K-1} \sum_{k \neq k^\star} \Big[C_{k^\star}^{\phi} - C_k^{\phi} + m\Big]_+ + w_{\text{s}} \left(\log \frac{\bar{C}_{\text{edge}}}{C_0}\right)^2. \tag{64}$$

The quality-KL term matches the learned cost-induced ranking to the throughput-derived target distribution, the cross-entropy term emphasizes the best candidate, the margin term separates the teacher path from alternatives, and the scale penalty prevents the learned edge costs from drifting to numerically large or small values without changing the ranking.

The edge-cost MLP uses three 512-dimensional hidden layers followed by a 256-dimensional projection and scalar output, resulting in 671,233 trainable parameters. The edge-cost GNN uses 128-dimensional node and edge embeddings, five message-passing layers, and a 128-dimensional output head, resulting in 548,865 trainable parameters. Both use learning rate 10^{-3}, at most 96 negative candidates per sample, margin $m = 1$, $T = 1$, $T_q = 0.25$, and weights $w_{\text{ce}} = 0.5$, $w_{\text{m}} = 0.25$, and $w_{\text{s}} = 0.01$. At inference, these models generate shortest-path candidates under the learned positive edge costs; the candidate is then committed using the same tree-preserving throughput evaluation used by the other graph schemes.

Appendix F
Sampling-Budget Analysis of the Numerical Upper-Bound Reference

We conducted a comparative analysis with a significantly larger trial size of 50,000 to validate that a numerical upper bound obtained from 5,000 random samples serves as a reliable benchmark. This validation is necessary because the reference itself is sampling-based, and its reliability depends on whether the trial budget is sufficiently large. As shown in Fig. 13, the 5,000- and 50,000-trial results remain very close, with only marginal differences across all settings. This suggests that 5,000 samples are sufficient to provide a stable and representative upper-bound estimate. Since the 50,000-trial setting incurs substantially higher computational overhead, we use 5,000 trials in the main experiments. We further confirmed this robustness by repeating both the 5,000- and 50,000-trial experiments 50 times for each data point, which again showed consistent trends.

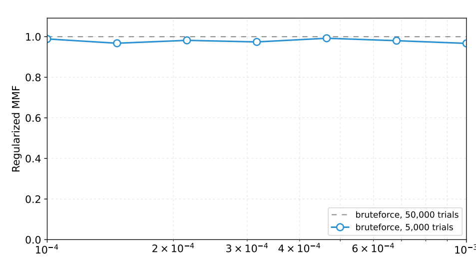

Fig. 13. Comparison of the **Numerical upper bound** scheme with 5,000 and 50,000 trials.

Appendix G
Implementation of the Message passing Algorithm

A. Routing Algorithm

Algorithm 2 Message passing relay routing

Require: Nodes $\mathcal{N}$

Ensure: Solution graph $\mathcal{G}^* = (\mathcal{N}^*, \mathcal{E}^*)$

$\mathcal{E}_\tau \leftarrow \{(i,j) \mid d^{\mathsf{s}}_{(i,j)} \leq D^{\mathsf{max}}_{(i,j)},\ i,j \in \mathcal{N}\}$, $\mathcal{N}^* \leftarrow \emptyset$, $\mathcal{E}^* \leftarrow \emptyset$

$\mathcal{G}_\tau \leftarrow (\mathcal{N}, \mathcal{E}_\tau)$, $\mathcal{G}^* \leftarrow (\mathcal{N}^*, \mathcal{E}^*)$, $T_{\mathcal{G}^*} \leftarrow 0$

Generate candidate nodes and paths $\mathcal{N}_{u,k}, \mathcal{E}_{u,k}$ for $u \in \mathcal{U}$ and $k = 1, \ldots, K$

repeat

  $T_{\text{old}} \leftarrow T_{\mathcal{G}^*}$

  for each user $u \in \mathcal{U}$ **do**

    $\mathcal{E}_{-u} \leftarrow \{(i,j) \in \mathcal{E}^* \mid \sum_{u' \in \mathcal{U}} x_{(i,j),u'} \neq x_{(i,j),u}\}$

    $\mathcal{N}_{-u} \leftarrow \{i, j \mid (i,j) \in \mathcal{E}_{-u}\}$

    for $k = 1, \ldots, K$ **do**

      $\mathcal{G}_{u,k} \leftarrow (\mathcal{N}_{-u} \cup \mathcal{N}_{u,k}, \mathcal{E}_{-u} \cup \mathcal{E}_{u,k})$

      $T_{\mathcal{G}_{u,k}} \leftarrow \min_{i \in \mathcal{I}} B_i \left(\sum_{j \in \mathcal{N}} \sum_{u' \in \mathcal{U}} \frac{x_{(i,j),u'} h_{u'}}{\gamma^*_{(i,j)}} \right)^{-1}$

    end for

    $\mathcal{G}^* \leftarrow \arg \max_{\mathcal{G} \in \{\mathcal{G}^*, \mathcal{G}_{u,1}, \ldots, \mathcal{G}_{u,K}\}} T_{\mathcal{G}}$, $(\mathcal{N}^*, \mathcal{E}^*) \leftarrow \mathcal{G}^*$

    $T_{\mathcal{G}^*} \leftarrow \max_{\mathcal{G} \in \{\mathcal{G}^*, \mathcal{G}_{u,1}, \ldots, \mathcal{G}_{u,K}\}} T_{\mathcal{G}}$

  end for

until $T_{\mathcal{G}^*} - T_{\text{old}} \leq \epsilon$

Message-passing structure. Message passing is the training-free path-replacement algorithm plotted as the message-passing baseline in the throughput-sweep figures. It is not a neural message-passing model. Instead, it exploits the tree structure of the routing graph: for each transmitter, the downstream hop counts and path-occupancy variables are aggregated from child nodes toward the source to evaluate the fixed-topology RRM objective in (27). When a user path is inserted, removed, or replaced, only the upstream branch affected by that user changes. The corresponding relay loads and throughput values can therefore be updated by backward propagation along that branch, while the rest of the tree keeps the same aggregated messages. Let $T_{\mathcal{G}}$ denote the max-min throughput of routing graph $\mathcal{G}$ after applying the closed-form RRM solution.

Routing procedure. Message passing first builds the SPSC-feasible super-graph $\mathcal{G}_\tau$ by applying the maximum link distance in (28). For each user, it constructs candidate paths from hop, distance, and randomized shortest-path predecessor lists. The randomized candidates are obtained by assigning independent random edge weights and extracting shortest paths on $\mathcal{G}_\tau$, which gives diverse feasible paths without enumerating all routes. The algorithm then refines the routing tree through sequential path replacement. For a selected user, Message passing temporarily deletes the user's current path, tests candidate reconnections that preserve the spanning-tree constraint in ($\mathcal{P}_1$c), updates only the affected upstream messages, and keeps the candidate with the largest analytical max-min throughput. If no candidate improves the objective in a round, the previous route is retained.

B. Computational Complexity

For U users, N nodes, K candidate paths per user, and R refinement rounds, shortest-path candidate generation costs $\mathcal{O}(KUN \log N)$, while refinement costs $\mathcal{O}(KUNR)$. The total complexity is $\mathcal{O}(KUN(\log N + R))$, or $\mathcal{O}(KUN \log N)$ when N dominates R.

C. Optimality Gap with Finite Candidate Paths

To quantify the effect of the finite candidate-path budget K, we define the throughput of a feasible routing graph $\mathcal{G}$ as

$$T_{\mathcal{G}} := \min_{\substack{u \in \mathcal{U}, \\ (i,j) \in \mathcal{E}_u}} B \left(\sum_{m \in \mathcal{N}} \sum_{u' \in \mathcal{U}} \frac{x_{(i,m),u'} h_{u'}}{\gamma^*_{(i,m)}} \right)^{-1}. \tag{65}$$

Let $\mathcal{G}^\star$ be a global maximizer of (27) and define $T^\star := T_{\mathcal{G}^\star}$. For each user $u \in \mathcal{U}$, let $\mathcal{E}^\star_u$ be the source-to-user path of u in $\mathcal{G}^\star$, and let $\mathcal{G}^\star_{-u} = (\mathcal{N}^\star_{-u}, \mathcal{E}^\star_{-u})$ denote the graph obtained from $\mathcal{G}^\star$ after removing $\mathcal{E}^\star_u$. For the k-th sampled path $\mathcal{E}_{u,k}$ in Alg. 2, define the single-user replacement regret as

$$\Delta_{u,k} := T^\star - T_{(\mathcal{N}^\star_{-u} \cup \mathcal{N}_{u,k},\, \mathcal{E}^\star_{-u} \cup \mathcal{E}_{u,k})}. \tag{66}$$

Also define the best regret among the K sampled paths as

$$\Delta_u^{(K)} := \min_{1\le k\le K} \Delta_{u,k}. \tag{67}$$

Let $F_u(t) := \Pr(\Delta_{u,1} \le t)$ be the cumulative distribution function of the one-sample regret and let

$$r_u(p) := \inf\{t \ge 0 : F_u(t) \ge p\}, \qquad p \in [0,1], \tag{68}$$

be its quantile function. Finally, let $\mathfrak{G}_K$ be the set of feasible routing graphs obtained by selecting one sampled path from $\{\mathcal{E}_{u,1}, \dots, \mathcal{E}_{u,K}\}$ for every user u while satisfying ($\mathcal{P}_1$c), and define the restricted optimum as

$$T_K^\star := \max_{\mathcal{G}\in\mathfrak{G}_K} T_{\mathcal{G}}. \tag{69}$$

Let $\widehat{\mathcal{G}}_K$ denote the output of Alg. 2, and define the local-refinement gap as

$$\xi_K := \mathbb{E}\Big[T_K^\star - T_{\widehat{\mathcal{G}}_K}\Big]. \tag{70}$$

Assumption 1 (Bounded interaction of single-user replacements)**.** *For every realization of the sampled candidate paths, the restricted optimum satisfies*

$$T^\star - T_K^\star \le \sum_{u\in\mathcal{U}} \Delta_u^{(K)}. \tag{71}$$

Theorem 2 (Finite-K routing gap of Message passing)**.** *Suppose that, for every user $u \in \mathcal{U}$, the K candidate paths in Alg. 2 are sampled independently and Assumption 1 holds. Then, the expected routing gap of Message passing satisfies*

$$\mathbb{E}\Big[T^\star - T_{\widehat{\mathcal{G}}_K}\Big] \le \sum_{u\in\mathcal{U}} \inf_{p\in[0,1]} \left\{r_u(p) + T^\star e^{-Kp}\right\} + \xi_K. \tag{72}$$

Moreover, if there exist constants $c_u > 0$, $\alpha_u > 0$, and $t_u > 0$ such that

$$F_u(t) \ge c_u t^{\alpha_u}, \qquad 0 \le t \le t_u, \tag{73}$$

then

$$\mathbb{E}\Big[T^\star - T_{\widehat{\mathcal{G}}_K}\Big] \le \sum_{u\in\mathcal{U}} \left[\frac{\Gamma\Big(1+\frac{1}{\alpha_u}\Big)}{(Kc_u)^{1/\alpha_u}} + T^\star e^{-Kc_u t_u^{\alpha_u}}\right] + \xi_K. \tag{74}$$

Proof. Fix $u \in \mathcal{U}$ and $p \in [0,1]$. By the definition of $r_u(p)$ in (68), we have $F_u(r_u(p)) \ge p$. Since the K sampled candidate paths are independent,

$$\begin{aligned}\Pr\Big(\Delta_u^{(K)} > r_u(p)\Big) &= \Pr(\Delta_{u,1} > r_u(p), \dots, \Delta_{u,K} > r_u(p)) \\ &= (1 - F_u(r_u(p)))^K \le (1-p)^K \le e^{-Kp}.\end{aligned} \tag{75}$$

Because $0 \le \Delta_u^{(K)} \le T^\star$, it follows that

$$\begin{aligned}\mathbb{E}\Big[\Delta_u^{(K)}\Big] &= \mathbb{E}\Big[\Delta_u^{(K)}\mathbf{1}\{\Delta_u^{(K)} \le r_u(p)\}\Big] + \mathbb{E}\Big[\Delta_u^{(K)}\mathbf{1}\{\Delta_u^{(K)} > r_u(p)\}\Big] \\ &\le r_u(p) + T^\star \Pr\Big(\Delta_u^{(K)} > r_u(p)\Big) \\ &\le r_u(p) + T^\star e^{-Kp}.\end{aligned} \tag{76}$$

Taking the infimum over $p \in [0,1]$ yields

$$\mathbb{E}\Big[\Delta_u^{(K)}\Big] \le \inf_{p\in[0,1]} \left\{r_u(p) + T^\star e^{-Kp}\right\}. \tag{77}$$

Next, by (70),

$$\mathbb{E}\Big[T^\star - T_{\widehat{\mathcal{G}}_K}\Big] = \mathbb{E}[T^\star - T_K^\star] + \xi_K. \tag{78}$$

Applying Assumption 1 and then (77), we obtain

$$\begin{aligned}\mathbb{E}\Big[T^\star - T_{\widehat{\mathcal{G}}_K}\Big] &\le \sum_{u\in\mathcal{U}} \mathbb{E}\Big[\Delta_u^{(K)}\Big] + \xi_K \\ &\le \sum_{u\in\mathcal{U}} \inf_{p\in[0,1]} \left\{r_u(p) + T^\star e^{-Kp}\right\} + \xi_K,\end{aligned} \tag{79}$$

which proves (72).

Now assume (73). For any $0 \le t \le t_u$,

$$\Pr\Big(\Delta_u^{(K)} > t\Big) = (1 - F_u(t))^K \le e^{-Kc_u t^{\alpha_u}}. \tag{80}$$

Using $\mathbb{E}[X] = \int_0^\infty \Pr(X > t)\, dt$ for a nonnegative random variable X, together with $\Delta_u^{(K)} \le T^\star$, gives

$$\begin{aligned}
\mathbb{E}\Big[\Delta_u^{(K)}\Big] &= \int_0^{T^\star} \Pr\Big(\Delta_u^{(K)} > t\Big)\, dt \\
&\le \int_0^{t_u} e^{-Kc_u t^{\alpha_u}}\, dt + T^\star e^{-Kc_u t_u^{\alpha_u}} \\
&\le \int_0^{\infty} e^{-Kc_u t^{\alpha_u}}\, dt + T^\star e^{-Kc_u t_u^{\alpha_u}} \\
&= \frac{\Gamma\Big(1 + \frac{1}{\alpha_u}\Big)}{(Kc_u)^{1/\alpha_u}} + T^\star e^{-Kc_u t_u^{\alpha_u}}.
\end{aligned} \tag{81}$$

Substituting (81) into (78) proves (74). $\square$

Remark 1. *If the one-sample regret distribution has a point mass at zero, i.e.,* $q_u := F_u(0) > 0$*, then* (72) *recovers the exact-optimal-support bound*

$$\mathbb{E}\Big[\Delta_u^{(K)}\Big] \le T^\star e^{-Kq_u}, \tag{82}$$

by choosing $p = q_u$.